AuthentiCheck: A Mobile-First Real-Time E-Commerce Fraud Auditor

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Introduction and Objectives

The problem

Counterfeit listings and manipulated reviews mislead online shoppers across mobile marketplaces.

AuthentiCheck delivers an on-demand live page audit for Amazon and Flipkart listings.

1 **Capture product URLs**

Native mobile share sheet intake or direct home-screen URL input.

2 **Prune live HTML**

Fetch live pages via ScraperAPI then strip non-semantic DOM nodes.

3 **Explain risk signals**

Generate deterministic JSON scores inside a high-contrast Evidence Ledger.

Comparative Literature Review

Study / Source	Core Focus	Identified Limitation	AuthentiCheck Approach
Bhangale & Roy (2025) [4]	LLM review analysis	High token overhead on raw pages	**Client DOM pruning before inference**
Harrison-Walker (2023) [5]	Five consumer cue families	Single review text lacks context	**Multi-signal Evidence Ledger**
JKSU Survey (2022) [6]	Supervised review classifiers	Labelled datasets become stale	**Zero-shot live listing evaluation**
Groq & Dart Docs (2026) [2,3]	Structured JSON schemas	Free-form text breaks mobile UI	**Strict AuthenticityResult parser**

AuthentiCheck integrates seller metadata, pricing anomalies, and review linguistics without relying on static central databases.

System Architecture

1. URL Intake

- Native OS Share Sheet
- Direct URL paste box
- Host scheme validation

2. Data Pipeline

- ScraperAPI proxy fetch
- Dart HTML5 DOM parser
- Semantic text extraction

3. AI Inference

- Groq Cloud LLM endpoint
- Fraud investigator prompt
- Strict JSON schema mode

4. Brutalist UI

- Live terminal audit logs
- Authenticity score 0–100
- Risk Factors Ledger

Serverless client-side orchestration connects ScraperAPI and Groq Cloud directly to the Flutter mobile dashboard.

On-Device DOM Pruning & Schema

>95%

Payload Reduction

Strips scripts and styles before LLM transmission.

3–4s

Target Audit Window

Fast end-to-end inference from URL intake to dashboard.

1. **Semantic evidence retention**
   Preserves product title, price, seller details, ratings, and review text.

2. **Strict JSON contract**
   Enforces authenticity_score, verdict, price_anomaly, and risk_factors fields.

3. **First-class failure handling**
   Surfaces explicit timeout or malformed JSON errors without silent fallbacks.

Modular Testing

Module	What is checked	Expected result
URL intake (TC-01, TC-02)	Share sheet capture and host validation	**Valid URLs accepted; invalid rejected**
Proxy scraping (TC-03, TC-04)	Live HTML retrieval via ScraperAPI	**HTML fetched or explicit error state**
DOM sanitization (TC-05)	Removal of scripts, styles, and menus	**Target semantic fields preserved**
JSON parsing (TC-06, TC-07)	Schema validation on Groq LLM output	**Score, verdict, and ledger rendered**
Terminal UI & flow (TC-09, TC-10)	Real-time progress logs and dashboard	**Smooth 3–4s end-to-end completion**

Implementation Status & Codebase

FLUTTER / DART MULTI-PLATFORM ARCHITECTURE | GROQ CLOUD & SCRAPERAPI

1

lib/main.dart

Initializes app state via flutter_dotenv.

2

home_screen.dart

Manages URL input and share intents.

3

scraper_service.dart

Fetches live HTML and prunes DOM.

4

llm_service.dart

Executes Groq Cloud heuristic prompts.

5

authenticity_result.dart

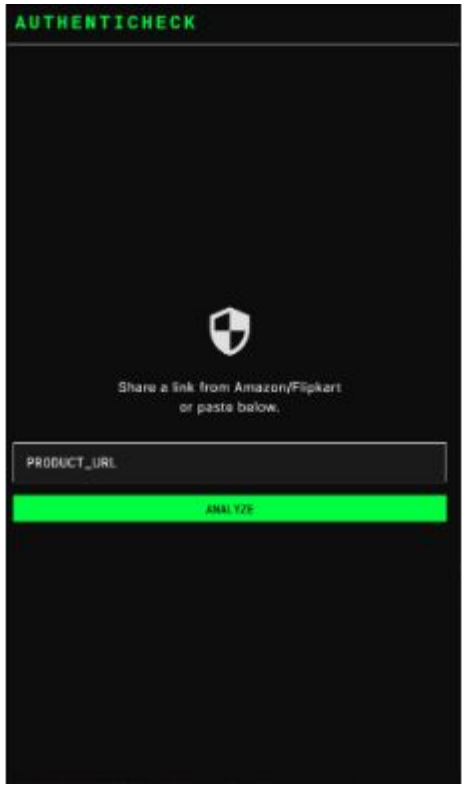
Validates structured JSON assessment data.

6

brutalist_theme.dart

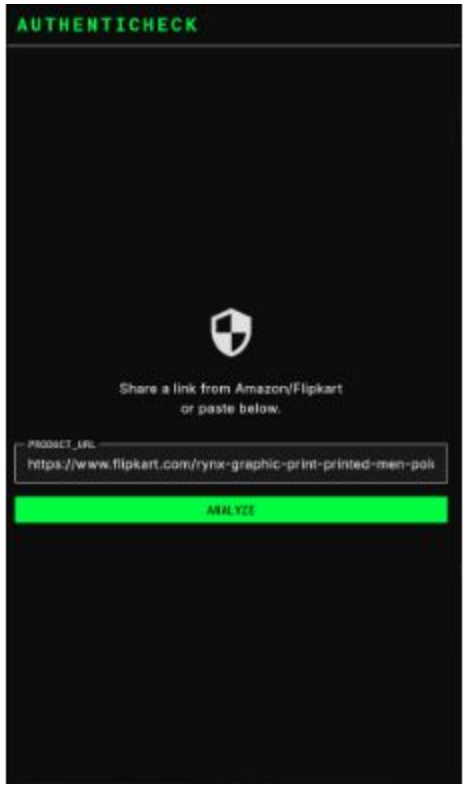
Defines high-contrast industrial UI styling.

Live Prototype: Intake & Proxy Scraping



1. Idle Home Screen

Ready for share intent or URL paste



2. Product URL Intake

Flipkart listing loaded into PRODUCT_URL



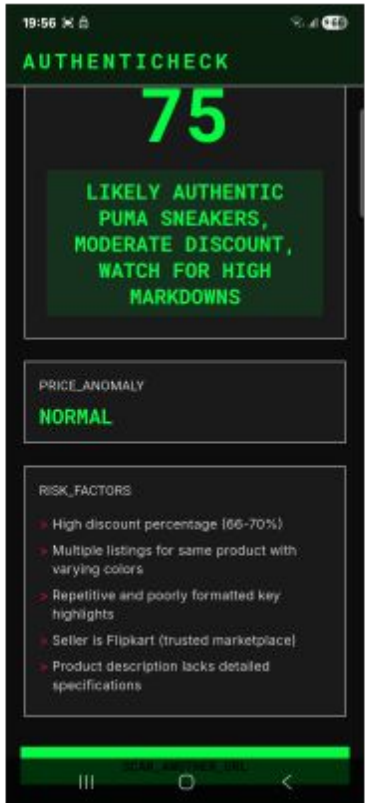
3. Live Proxy Scraping

Terminal log: Fetching from ScraperAPI

Results: Live Audit & Evidence Ledger



4. LLM Heuristic Execution



5. Rendered Evidence Ledger

Score: 75 / 100

PRICE_ANOMALY: NORMAL

- High discount percentage (66–70%)
- Multiple color variant listings
- Repetitive key highlight formatting
- Trusted Flipkart marketplace seller
- Missing detailed product specifications

Conclusion and Future Scope

Conclusion & Key Takeaways

- Evaluates live e-commerce listings at purchase time without stale databases.
- Reduces HTML payload size over 95% via on-device DOM pruning.
- Surfaces transparent risk factors inside a deterministic JSON Evidence Ledger.

Scalability & Future Scope

- Multi-marketplace adapters for regional storefronts and historical price tracking.
- Computer vision checks for logo mismatches and counterfeit packaging.
- Local SQLite scan history and calibrated confidence scoring layers.

THANK YOU !!

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